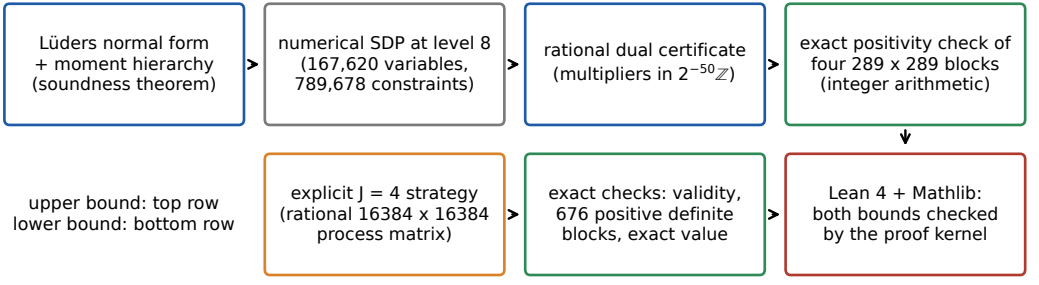
Lüders normal form
+ moment hierarchy
(soundness theorem)
numerical SDP at level 8
(167,620 variables,
789,678 constraints)
rational dual certificate
(multipliers in $2^{-50}\mathbb{Z}$)
exact positivity check of
four 289 x 289 blocks
(integer arithmetic)
upper bound: top row
lower bound: bottom row
explicit J = 4 strategy
(rational 16384 x 16384
process matrix)
exact checks: validity,
676 positive definite
blocks, exact value
Lean 4 + Mathlib:
both bounds checked
by the proof kernel